

# 1 Jarvis and Mckenzie, 1980 General equations

Conservation of mass, momentum and energy:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (1)$$

$$\rho \left( \frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla P + \rho g + \frac{\partial}{\partial x_j} \tau_{ij} \quad (2)$$

$$\rho C_p \left\{ \frac{\partial T}{\partial t} - \frac{\alpha T}{\rho C_p} \frac{\partial P}{\partial t} + \mathbf{u} \cdot [\nabla T - (\nabla T)_s] \right\} = \nabla \cdot (K \nabla T) + H + \tau_{ij} \frac{\partial u_i}{\partial u_{ij}} \quad (3)$$

All terms and symbols used in these equations are shown in table 1. The (Newtonian) relation between the deviatoric stress tensor  $\tau_{ij}$  and the velocity field is

$$\tau_{ij} = \eta \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \delta_{ij} \frac{\partial u_k}{\partial x_k} \right) + \eta_2 \delta_{ij} \frac{\partial u_k}{\partial x_k}. \quad (4)$$

Combining 4 with eq. 3 (momentum) gives the general Navier-Stokes equations, but Jarvis and Mckenzie (1980) does not show these. Instead, they assume that  $\frac{1}{Pr} \rightarrow 0$ , which they assume (do they?) is the case for the Earth's mantle, the inertial forces can be ignored. The resulting equations will be discussed in the next section, which covers the approximations made by Jarvis and Mckenzie (1980).

## 2 Jarvis and Mckenzie, 1980 Approximations

Problem statement: for the Boussinesq approximation to be valid, the depth range of convection  $d$  must be small compared to the temperature scale height  $H_T$  ( $d/H_T \ll 1$ ). This is not the case for many systems of deep convection, amongst which the convection of planetary mantles. Physically,  $d/H_T \approx O(1)$  means that the influence of viscous dissipation, variable adiabatic temperature- and density gradients must be maintained in the set of governing equations.

The assumptions made for the **anelastic liquid approximation** are that  $d = O(H_T)$  and that  $\alpha T \ll 1$  **what does this mean and how does this relate to Prandtl number? and if  $\alpha \ll 1$ , how come  $D$  is not  $\ll 1$ ?**

### 2.1 Reference state and equation of state

A reference state with hydrostatic pressure  $P_H$  and adiabatic temperature distribution (gradient?)  $T_s$  is considered. The equation of state is written as a linear Taylor expansion of  $\rho$  about the reference state density distribution  $\rho_r$ :

$$\rho(T, P) = \rho_r [1 - \alpha(T - T_s) + K_T^{-1}(P - P_H)] \quad (5)$$

Where

$$\alpha = \frac{1}{\rho_r} \left( \frac{\partial \rho_r}{\partial T} \right)_P \quad (6)$$

$$K_T = \rho_r \left( \frac{\partial P}{\partial \rho_r} \right)_T \quad (7)$$

Determining the reference density distribution  $\rho_r$  is done by considering the gradient of  $\rho_r(S, P_H)$  with  $S$  the entropy per unit mass:

$$\nabla \rho_r(S, P_H) = \left( \frac{\partial \rho_r}{\partial S} \right)_{P_H} \nabla S + \left( \frac{\partial \rho_r}{\partial P_H} \right)_S \nabla P_H \quad (8)$$

With  $\nabla S = 0$  since the reference state is adiabatic, and with  $\nabla P_H = -\rho_r g \hat{\mathbf{z}}$ :

$$\frac{1}{\rho_r} \frac{\partial \rho_r}{\partial z} = -\frac{\rho_r g}{K_s} \quad (9)$$

with  $K_s = \rho_r \left( \frac{\partial P_H}{\partial \rho_r} \right)_S$ . Using the definition of the Gruneisen parameter  $\Gamma$  and the temperature scale height of the liquid  $H_T^s$  gives the Adams-Williamson relation

$$\frac{1}{\rho_r} \frac{\partial \rho_r}{\partial z} = -\frac{1}{H_T \Gamma} \quad (10)$$

Which is valid for a **chemically homogeneous adiabatic density distribution under hydrostatic pressure**. Integration of equation 10 over  $z$  gives the reference density distribution as

$$\rho_r(z) = \rho_0 \exp[(d - z)/H_T \Gamma] \quad (11)$$

where  $\rho_0 = \rho_r(z = d)$  is the reference density value at the surface **orientation of coordinate system??**. Plugging this expression of the reference density distribution into 5 gives the (linearized equation) of state

$$\rho(T, P) = \rho_0 \exp[(d - z)/H_T \Gamma] [1 - \alpha(T - T_s) + K_T^{-1}(P - P_H)] \quad (12)$$

## 2.2 Nondimensionalization

2, 3. **move tables here?**

## 2.3 The anelastic-liquid approximation

From now on, only the dimensionless variables will be used but with primes suppressed (i.e. all variables are assumed dimensionless). The dimensionless equation of state (from eq. 12) becomes

$$\rho = \rho_r [1 - \mu(T - T_s) + \mu(\rho_0 g d / K_T) P_1] \quad (13)$$

with  $\mu = \alpha T_1$ . If  $\mu \rightarrow 0$ ,  $\rho$  reduces to its reference profile  $\rho_r$  (as is the case in the Boussinesq approximation).

## 3 Appendix 1: tables of symbols

Table 1: Symbols used by Jarvis and Mckenzie, 1980

|                |                                                             |                                                    |
|----------------|-------------------------------------------------------------|----------------------------------------------------|
| $d$            |                                                             | depth of the convecting layer                      |
| $H_T$          | $\frac{C_p}{g\alpha}$                                       | temperature scale height of the liquid             |
| $C_p$          |                                                             | specific heat at constant pressure                 |
| $C_v$          |                                                             | specific heat at constant volume                   |
| $\alpha$       |                                                             | volume coefficient of thermal expansion            |
| $\tau_{ij}$    |                                                             | deviatoric stress tensor                           |
| $(\nabla T)_s$ | $\frac{\alpha T}{\rho C_p} \nabla P$                        | adiabatic temperature gradient                     |
| $K$            |                                                             | thermal conductivity                               |
| $H$            |                                                             | rate of internal heat production                   |
| $\eta$         | $\rho\nu$                                                   | dynamic viscosity                                  |
| $\nu$          |                                                             | kinematic viscosity                                |
| $\eta_0$       | $\ll 1$                                                     | second or bulk viscosity                           |
| $\Gamma$       | $\frac{\alpha K_s}{\rho C_p} = \frac{\alpha K_T}{\rho C_v}$ | Gruneisen's parameter, assumed constant and $O(1)$ |
| $K_s$          |                                                             | isentropic modulus of bulk compressibility         |
| $K_T$          |                                                             | isothermal modulus of bulk compressibility         |
| $\mu$          | $\alpha T_1$                                                |                                                    |
| $\alpha$       | $10^{-5} \text{ } ^\circ K^{-1}$                            | volume coefficient of thermal expansion            |

Table 2: Characteristic values of Jarvis and Mckenzie, 1980

|             |                                                                |
|-------------|----------------------------------------------------------------|
| depth       | $d$                                                            |
| temperature | $T_1 = \frac{1}{K} \text{WHAT??} \rightarrow 1^\circ K$        |
| density     | $\rho_0$                                                       |
| velocity    | $U = \frac{g\alpha T_1 d^2}{\nu_0}, \quad \nu_0 = \eta/\rho_0$ |
| time        | $\tau = d/U$                                                   |

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Table 3: Dimensionless variables in Jarvis and Mckenzie, 1980

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|               |                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------|
| $(x', z')$    | $(x, z)/d$                                                                                             |
| $\rho'_r$     | $\rho_r/\rho_0 = \exp[(1 - z')D/\Gamma], \quad D = d/H_T$                                              |
| $\rho'$       | $\rho/\rho_0$                                                                                          |
| $T'$          | $(T - T_0)/T_1$                                                                                        |
| $t'$          | $t/\tau$                                                                                               |
| $P'_1$        | $\frac{P_1 d}{U\eta} = P_1 \alpha T_1 \rho_0 g d, \quad P_1 = P - P_H$                                 |
| $\kappa'_0$   | $\kappa_0 \nu_0 / g \alpha T_1 d^3, \quad \text{thermal diffusivity}$                                  |
| $\epsilon'_0$ | $\epsilon_0 \nu_0 / g \alpha T_1^2 d, \quad \text{energy generation rate, } \epsilon_0 = H/\rho_0 C_p$ |
| $\lambda$     | $L/d, \quad \text{box width, } L \text{ is the dimensional box width}$                                 |

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